

POCKETGULL HEALTH AI PBC

CLINICAL ARCHITECTURE • BIOPHYSICAL SIMULATION • ORTHOBIOLOGICS
TRANSLATION

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STATUS: Confidential Collaboration
Memorandum

TO THE EXECUTIVE LEADERSHIP OF:

Dr. Robert E. Guldberg, Vice President & Leona DeArmond Executive Director, Knight Campus
Dr. Benjamín J. Alemán, Department of Physics, Alemán Lab & CAMCOR
Dr. Paul Dalton, Department of Bioengineering, Biofoundry & VAM Specialist
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1. EXECUTIVE THESIS: THE ASYMMETRIC ADVANTAGE

Tendon and ligament reconstruction remains tethered to a stagnant standard of care: **cadaver allografts and inert synthetic meshes**. While institutions like Harvard Wyss, Wake Forest (WFIRM), and the University of Washington invest heavily in generic layer-by-layer extrusion bioprinting, they face severe translational hurdles: slow printing speeds, high cell mortality, and destructive end-point assays that make continuous quality control impossible.

The University of Oregon possesses an unprecedented **triad of asymmetric assets** located in immediate geographic proximity:

- Volumetric Additive Manufacturing (VAM)** at the Knight Campus Biofoundry, curing full 3D living tissue constructs in **30–60 seconds** (100× faster than extrusion);
- Suspended 2D Graphene Nanoresonators ("Graphene Trampolines")** at the Alemán Physics Lab and CAMCOR, delivering non-destructive real-time mass shift (Δf) and piconewton cellular traction force telemetry (pN);
- Hayward Field and the Bowerman Sports Science Clinic**, generating the highest empirical musculoskeletal ground-reaction forces (>9,000 N) in human athletics.

2. COMPARATIVE BENCHMARKING: LIVING VS. STATIC SUBSTITUTES

Biomechanical Metric	Cadaver Allograft (Standard)	Synthetic Polymer (PLLA/Dacron)	UO Knight Living Construct (VAM)
Revascularization & Ingress	Slow / Incomplete (6–18 mos)	Zero (Scar encapsulation)	Rapid (14–21 days via 150 μ m pores)

Biomechanical Metric	Cadaver Allograft (Standard)	Synthetic Polymer (PLLA/Dacron)	UO Knight Living Construct (VAM)
Cyclic Creep & Fatigue	High failure risk under sprint loads	Permanent deformation creep	Active remodeling (continuous repair)
Real-Time Viability Verification	Destructive biopsy only	Not applicable	Alemán Graphene Resonance (Δf , pN)
Elite Athlete Rerupture Rate	12% – 18%	High long-term hazard	Estimated <3% (physiologically adapted)

3. PROPOSED TURNKEY COLLABORATION DELIVERABLES

PocketGull LLC (incorporated in Oregon, geofenced natively to the us-west1 Oregon Local Zone) has completed the reference implementation for three core digital capabilities, available to the University of Oregon at zero software licensing cost:

- **In-Silico WebGL Digital Twin:** High-speed parameter sweep engine (Krogh diffusion, pore size, cyclic strain) eliminating up to 80% of experimental wet-lab material waste before printing.
- **Bio-Graphene Ingestion Pipeline:** Real-time telemetry service capturing Sauerbrey mass deposition and cellular traction forces with NIST SP 800-90A CSPRNG entropy.
- **FDA 21 CFR Part 11 Regulatory Dossier:** Automated cryptographic SHA-256 tamper-evident audit ledger and HIPAA §164.514 Safe Harbor de-identification pipeline ready for OHSU/OCTRI animal trials.

4. NEXT STEPS

We propose an initial 30-minute virtual alignment session with your bioengineering and innovation teams to demonstrate the interactive digital twin and discuss provisional patent coordination.